Question Paper 1

Part – A (Mandatory to answer)

CLO1 1. a. Define interrupt and TRAP (2)

CLO2 b. Suppose you have opened a memorable image in your computer. Explain the usage of PCB regarding the given case with a diagram. (4)

CLO3 c. Suppose that a disk drive has 5000 cylinders, numbered 0 to 4999. The drive is currently serving a request at cylinder 143, and the previous request was at cylinder 125. The queue of pending requests, in FIFO order, is (6)

86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130

Starting from the current head position, what is the total seek distance(in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk-scheduling algorithms?

i. FCFS ii. SSTF iii. SCAN iv. LOOK v. C-SCAN vi. C-LOOK

2. a. Rank the following storage systems from slowest to fastest with a diagram: (3)

a. Hard-disk drives b. Registers c. Optical disk d. Main memory e. Nonvolatile memory f. Magnetic tapes g. Cache

b. Define each part of a process. (2)

With a diagram show the memory layout of the given C program in an Operating System. (+ 4)

```
#include <stdio.h>
#include <stdlib.h>
int global_var = 10;
int uninit_global;
void function() {
    int local_var = 5;
    printf("Inside function: local_var = %d\n", local_var);
}
int main() {
    static int static_var = 20;
    int local_main = 15;
    int *ptr = malloc(sizeof(int));
    *ptr = 25;
    printf("global_var: %d\n", global_var);
    printf("uninit_global: %d\n", uninit_global);
    printf("static_var: %d\n", static_var);
    printf("local_main: %d\n", local_main);
    printf("*ptr (heap): %d\n", *ptr);
```

```
        function();
        free(ptr);
        return 0;
    }
```

c. Discuss about file-system and mass-storage management of OS. (3)

3. a. Can a multithreaded solution using multiple user-level threads achieve better performance on a multiprocessor system than on a single-processor system? Explain. (3)

b. Which of the following components of program state are shared across threads in a multithreaded process? Show with a diagram. (2)

a. Register values b. Heap memory c. Global variables d. Stack memory

c. Discuss about the types of parallelism. (4)

d. Determine whether the following problems exhibit task or data parallelism: (3)

- Using a separate thread to generate a thumbnail for each photo in a collection
- Transposing a matrix in parallel
- A networked application where one thread reads from the network and another writes to the network.

4. a. Explain process scheduling and how it maintains different scheduling queues of processes? Represent process scheduling using queueing diagram. (3)

b. How does process cooperation enhance convenience for users? Differentiate between message-passing and shared-memory models of IPC. (3)

c. Consider the following set of processes, with the length of the CPU burst time given in milliseconds: (6)

Process	Burst Time	Priority
P1	2	2
P2	1	1
P3	8	4
P4	4	2
P5	5	3

The processes are assumed to have arrived in the order P1, P2, P3, P4, P5, all at time 0.

i. Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, non-preemptive priority (a larger priority number implies a higher priority), and RR (quantum=2).

ii. What is the turnaround time of each process for each of the scheduling algorithms?

iii. What is the waiting time of each process for each of these scheduling algorithms?

iv. Which of the algorithms results in the minimum average waiting time (over all processes)?

5. a. Discuss about producer consumer problem. (2)

b. Illustrate critical section problem with a diagram. (4)

c. Elaborate reader-writers problem with its variations. (6)

6. a. Explain deadlock with an example using two semaphores. (3)

Consider the following snapshot of a system:

Process	Allocation (A B C D)	Max (A B C D)	Available (A B C D)
T0	3 1 4 1	6 4 7 3	2 2 2 4
T1	2 1 0 2	4 2 3 2	
T2	2 4 1 3	2 5 3 3	
T3	4 1 1 0	6 3 3 2	
T4	2 2 2 1	5 6 7 5	

Answer the following questions using the banker's algorithm:

b. Illustrate that the system is in a safe state by demonstrating an order in which the threads may complete. (3)

c. If a request from thread T4 arrives for (2,2,2,4), can the request be granted immediately? (3)

d. If a request from thread T2 arrives for (0,1,1,0), can the request be granted immediately? (3)

7. a. Briefly explain the memory management unit with a diagram. (4)

b. Illustrate variable partitioning with a diagram. (3)

c. Explain two types of fragmentation. (2)

d. Consider a logical address space of 128 pages with a 1024 bytes page size, mapped onto a physical memory of 62 page frames. (3)

a. How many bits are required in the logical address?

b. How many bits are required in the physical address?

Question Paper 2

(CLO1) 1. (a) Explain Bootstrap program using figure. Identify several advantages and disadvantages of open-source operating systems. 3

(CLO2) (b) How does an interrupt differ from a trap? Discuss the interrupt-driven I/O cycle with figure. 3

(CLO3) (c) Suppose that a disk drive has 5000 cylinders, numbered 0 to 4999. The drive is currently serving a request at cylinder 143, and the previous request was at cylinder 125. The queue of pending requests, in FIFO order, is 6

86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130

Starting from the current head position, what is the total seek distance(in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk-scheduling algorithms?

i. FCFS ii. SSTF iii. SCAN iv. LOOK v. C-SCAN vi. C-LOOK

Part: B

Note: Answer any Four (4) of the following questions from Part B. Mark of each question are mentioned at the right margin respectively.

2. (a) Explain process scheduling and how it maintains different scheduling queues of processes? Represent process scheduling using queueing diagram. 3

(b) How does process cooperation enhance convenience for users? Differentiate between message-passing and shared-memory models of IPC. 3

(c) Consider the following set of processes, with the length of the CPU burst time given in milliseconds: 6

Process	Burst Time	Priority
P1	2	2
P2	1	1
P3	8	4
P4	4	2
P5	5	3

The processes are assumed to have arrived in the order P1, P2, P3, P4, P5, all at time 0.

i. Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, non-preemptive priority (a larger priority number implies a higher priority), and RR (quantum=2).

ii. What is the turnaround time of each process for each of the scheduling algorithms?

iii. What is the waiting time of each process for each of these scheduling algorithms?

iv. Which of the algorithms results in the minimum average waiting time (over all processes)?

3. (a) What are the benefits of using threads in a multithreaded environment? Discuss the disadvantages of user threads compared to kernel threads. 3

(b) How do you determine deadlock in a resource allocation graph? Does the following resourse allocation graph contain a deadlock? Justify your answer. 3

(Diagram containing nodes P1, P2, P3 and resource nodes R1, R2, R3)

(c) Assume that there are 5 processes, P0 through P4, and 4 types of resources. At T0 we have the following system state: 6

Max Instances of Resource Type A = 3 (2 allocated + 1 Available)

Max Instances of Resource Type B = 17 (12 allocated + 5 Available)

Max Instances of Resource Type C = 16 (14 allocated + 2 Available)

Max Instances of Resource Type D = 12 (12 allocated + 0 Available)

Process	**Allocation A**	**B**	**C**	**D**	**Max A**	**B**	**C**	**D**	**Available A**	**B**	**C**	**D**
P0	0	1	1	0	0	2	1	0	1	5	2	0
P1	1	2	3	1	1	6	5	2				
P2	1	3	6	5	2	3	6	6				
P3	0	6	3	2	0	6	5	2				
P4	0	0	1	4	0	6	5	6				
Total	2	12	14	12								

i. Use the safety algorithm to test if the system is in a safe state or not?

ii. If the system is in a safe state, can the following requests be granted, why or why not? Please also run the safety algorithm on each request as necessary.

a. P1 requests (2,1,1,0)

b. P1 requests (0,2,1,0)

4. (a) Consider following page reference string: 7,0,1,2,0,3,0,4,2,3,0,3,0,3,2,1,2,0,1,7,0,1. Compare the performance for the following page replacement algorithms assuming three frames. 4

i. Optimal Algorithm

ii. Least Recently Used Algorithm

(b) Define page fault? Discuss the steps in handling a page fault with figure? What Happens if there is no Free Frame? 4

(c) Mention the causes of thrashing? How does the system detect thrashing? 4

Considering the following segment table and page table, translate the following logical address to physical address: (2,6), where page size=2.

Segment	Limit	Page Table Base
S0	10	1000
S1	14	1200
S2	12	1400

P0	1000	F3
P1	1001	F2
P2	1002	F1
P3	1003	F0

5. (a) Show with an example how the size of main memory increases the CPU utilization? 3

(b) Compare internal fragmentation with external fragmentation. To minimize the probability of external fragmentation what we can use? Discuss the process. 3

(c) How would first-fit and best-fit algorithms place the following set of processes in order? 6

Process	Size	Block	Size
P1	100k	B1	50k
P2	10k	B2	200k
P3	35k	B3	70k
P4	15k	B4	115k
P5	23k	B5	15k
P6	6k		
P7	25k		
P8	55k		
P9	88k		
P10	100k		

6. (a) List three examples of deadlocks that are not related to a computer system environment. 3

(b) Describe briefly the process of recovery from deadlock by process termination and resource preemption. 3

(c) Consider a logical address space of 64 pages of 1,024 words each, mapped onto a physical memory of 32 frames. 6

a. How many bits are there in the logical address?

b. How many bits are there in the physical address?

7. (a) How address binding of instructions and data to memory addresses can happen? 2

(b) OS is responsible for the protection of user's memory. This is accomplished by hardware. Draw the hardware and describe its operation principle. 6

(c) Describe the general dynamic storage allocation problem which concerns how to satisfy a request of size n from a list of free holes. 4

Question Paper 3

1. a) "Operating system works as a resource allocator as well as program controller" - Justify the statement. 2.0

b) How does the direct memory access (DMA) technique work? Explain with a proper diagram. 3.0

c) Briefly describe the Dual-mode operation of an operating system. 3.0

d) Define system call. Suppose you want to copy the contents of an existing file to a new file. Explain the process with a system call. 4.0

2. a) How address binding of instructions and data to memory addresses can happen? 4.0

b) OS is responsible for the protection of the user's memory. This is accomplished by hardware. Draw the hardware and describe its operation principle. 4.0

c) Describe the general dynamic storage allocation problem which concerns how to satisfy a request of size n from a list of free holes. 4.0

3. a) Write the types of execution that concurrent processes can perform. 3.0

b) What are the benefits of using cooperating process? How Interprocess communication is performed? 4.0

c) Write the solution goals of critical section problems in process synchronization. Mr. Mahmud has proposed a solution for process synchronization to access the critical section. Justify the solution goals and problems of the solution (if any): 5.0

```
bool LOCK = false; // shared


while (true)
{
   // do non_critical_section
   while (LOCK == true) {}
   LOCK = true
   // do critical_section


   LOCK = false
}
```

4. a) Consider the following set of processes with the CPU burst time, arrival time, and priority. 9.0

Process	Arrival Time	Burst Time	Priority
P1	5	8	4
P2	0	5	1
P3	2	3	2
P4	0	3	5

Process	Arrival Time	Burst Time	Priority
P5	2	2	3

Draw three Gantt charts illustrating the execution of the processes using the following algorithms. Also, calculate the average waiting time and average turnaround time for each of the algorithms.

i. Preemptive Priority Scheduling

ii. Preemptive Shortest Job First

iii. Round Robin with Time Quantum=3

b) Define the starvation problem and the convoy effect. How these can be solved? 3.0

5. a) How do you determine deadlock in a resource allocation graph? Does the following resource allocation graph contain a deadlock? Justify your answer. 4.0

(Diagram containing process nodes P1, P2, P3 and resource nodes R1, R2, R3)

b) Considering the following snapshot of a system, answer the questions using the banker's algorithm: 8.0

Process	Allocation A	B	C	D	Max A	B	C	D	Available A	B	C	D
P0	0	0	1	2	0	0	1	2	1	5	2	0
P1	1	0	0	0	1	7	5	0				
P2	1	3	5	4	2	3	5	6				
P3	0	6	3	2	0	6	5	2				
P4	0	0	1	4	0	6	5	6				

i) Find the total number of instances in the system for each of the resources.

ii) What is the content of the need matrix?

iii) Is the system in a safe state? If yes, find the safe sequence.

iv) If a request form process P1 arrives for (0, 4, 2, 3), can the request be granted immediately? Justify your answer.

6. a) Consider the following page reference string: 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3. Compare the performance for the following page replacement algorithms assuming three frames. 4.0

i. Least Recently Used Algorithm

ii. Optimal Algorithm

b) How would first-fit and best-fit algorithms place the following set of processes in order? 6.0

Process	Size	Block	Size
P1	212 k	B1	100 k

Process	Size	Block	Size
P2	417 k	B2	500 k
P3	112 k	B3	200 k
P4	426 k	B4	300 k

c) Consider the following segment table: 2.0

Segment	Base	Limit
0	219	600
1	2300	14
2	90	100
3	1327	580
4	1957	96

What are the physical addresses for the following logical addresses?

i) 0, 430 ii) 1, 10 iii) 2, 250 iv) 4, 112

7. a) Define fragmentation. Compare internal fragmentation with external fragmentation. 3.0

b) Suppose that a disk drive has 2500 cylinders numbered 0 to 2499. The drive is currently serving a request at cylinder 2150. The queue of pending requests in order is: 6.0

20, 69, 212, 2, 296, 800, 544, 618, 356, 523, 965, 36, 81

Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the requests for the following disk scheduling algorithms:

(i) SCAN (ii) C-LOOK (iii) SSTF

c) What is the purpose of disk scheduling? You have to select an algorithm for disk scheduling. Identify the criteria you will follow for selecting a proper disk scheduling algorithm. 3.0

Question Paper 4

1. a. Illustrate the relationships among system & application program, operating system and computer hardware. (3)

b. How does an open source development model work? "Linux is an open source operating system" – Justify the statement. (3)

c. Define dual-mode operation of an operating system. Show the transition from user to kernel mode in dual mode operation with an example. (3)

d. Suppose you want to move the contents of an existing file to a new file. Explain the process with system call sequence. (3)

2. a. What is context switch? Show with pictorial representation of Context switching. Give the brief descriptions of multilevel-feedback queue. (4)

b. Cooperating processes need inter-process communication. What are two models of inter-process communication? Describe. (4)

c. What are the benefits of using multi-threading instead of multiple process? Differentiate between the user thread and kernel thread. (4)

3. a. Consider the scenario of the read and write problems of operating system, one program writes and the other reads, and read and write don't interfere with each other. Design a solution to the problem using Semaphores ensuring the following conditions: (5)

   i. If there is a read operation inside the critical section, no other read or write inside the critical section.

   ii. If there is a write operation inside the critical section, no other read or write inside the critical section.

b. Analyze the Peterson's solution to decide whether it provides all the necessary requirements of synchronization mechanisms or not. (4)

c. Criticize the Lock Variable mechanism of synchronization. (3)

4. a. CPU scheduling decisions take place when a process changes its states. Identify the changes of states when CPU scheduling is necessary. Also, classify them as preemptive and non-preemptive scheduling. (3)

b. Three process P1, P2 and P3 arrive at time zero. The total time spent by the process in the system is 10ms, 20ms, and 30ms respectively. They spent first 20% of their execution time in doing I/O and the rest 80% in CPU processing. What is the percentage utilization of CPU using FCFS scheduling algorithm? (3)

c. Consider the set of 4 processes whose arrival time and burst time are given below– (6)

Process	Burst Time	Arrival Time
P1	6	2
P2	2	5
P3	8	1

Process	Burst Time	Arrival Time
P4	3	0
P5	4	4

If the CPU scheduling policy is Shortest Remaining Time First (SRTF), calculate the average waiting time and average turnaround time.

5. a. For the following snapshot, draw the gantt chart using EDF algorithm (5)

Process	Period	Execution Time
P1	10	5
P2	20	6
P3	15	3

Find the total utilization of the system.

b. Assume that we have following resources: 5 pendrives, 2 printers, 4 scanners & 3 disks. The resources are allocated as per the matrix below: (5)

Process	Alloc: pendrive	printer	scanner	disk	Max: pendrive	printer	scanner	disk
P1	2	0	1	1	1	1	0	0
P2	0	1	0	0	0	1	1	2
P3	1	0	1	1	2	1	0	0
P4	1	1	0	1	0	0	1	0

i) Find the available number of instances in the system for each of the resources.

ii) What is the content of the max matrix?

Is the system in a safe state? If yes, find the safe sequence.

c. Illustrate safety algorithm and resource-request algorithm with appropriate flowcharts. (3)

6. a. Given five memory partitions of 100KB, 500KB, 200KB, 300KB, and 600KB. How would the first-fit, best-fit, and worst fit algorithms place processes of 212KB, 417KB, 112KB, and 426KB? Which algorithm makes the most efficient use of memory? (6)

b. Considering the following segment table and page table, translate the following logical address to physical address: (2,6), where page size=2. (6)

Segment	Limit	Page Table Base
S0	10	1000

Segment	Limit	Page Table Base
S1	14	1200
S2	12	1400

P0	1000	F3
P1	1001	F2
P2	1002	F1
P3	1003	F0

7. a. Define Hit ratio. Assume that a 90% hit ratio in the TLB 80% of the time and 10 ns is needed to memory. Find the EAT. (4)

b. Calculate the size of memory if its address consists of 22 bits and memory is addressable. (4)

c. Consider a machine with 64 MB physical memory and a 32-bit virtual address space. If the page size is 4KB then determine the approximate size of the page table? (4)